



Introduction to IT 2025

L2

Introduction to LaTeX

Learn LaTeX in 30 minute

What is LaTeX?

- LaTeX (pronounced "LAY-tek" or "LAH-tek") is a tool for typesetting professional-looking documents.
- Key difference:
 - Not WYSIWYG like Word
 - Works with plain text files containing LaTeX commands
 - TeX engine processes and produces professional PDF files
- Philosophy: You focus on content, LaTeX handles the presentation!

Why Learn LaTeX?

Advantages:

- Excellent support for complex mathematics and equations.
- Facilities for footnotes, cross-referencing, and bibliography management.

Why Learn LaTeX?

Advantages:

- Automated generation of table of contents, lists of figures, indexes.
- Highly customizable through thousands of free packages.
- Separation of content and style.

Your First LaTeX Document

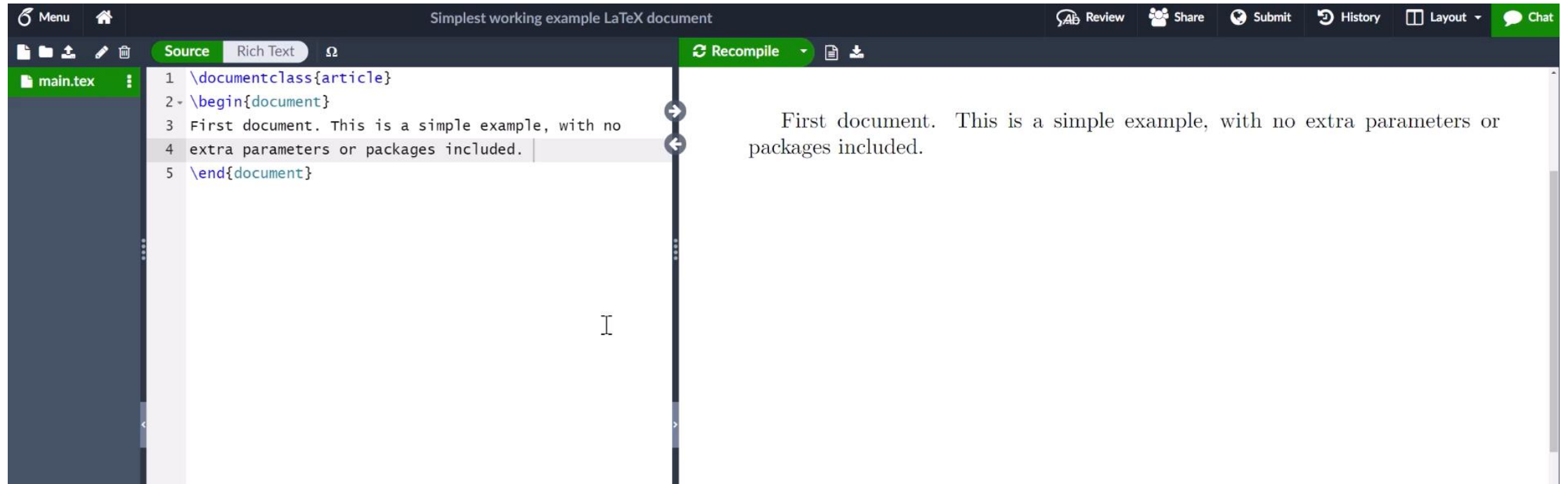
```
\documentclass{article}
```

```
\begin{document}
```

First document. This is a simple example, with no extra parameters or packages included.

```
\end{document}
```

Your First LaTeX Document



Your First LaTeX Document

Menu Review Share Submit History Layout Chat

Simplest working example LaTeX document

Source Rich Text Ω

main.tex

```
1 \documentclass{article}
2 \begin{document}
3 First document. This is a simple
  example, with no extra parameters or
  packages included. Now I have added
  more text and need to recompile the
  document.
4 \end{document}
5
```

File outline

We can't find any sections or subsections in this file. [Find out more about](#)

Recompile

- Auto Compile
 - On
 - ✓ Off
- Compile Mode
 - ✓ Normal
 - Fast [draft]
- Syntax Checks
 - ✓ Check syntax before compile
 - Don't check syntax
- Compile Error Handling
 - Stop on first error
 - ✓ Try to compile despite errors
- Stop compilation
- Recompile from scratch

Select the downward-facing arrow

Set "Auto Compile" to "On" or "Off"

Document Structure: The Preamble

- **Preamble** is the "setup" section before `\begin{document}`.
- In the Preamble, you can: Configure document class; Load packages; Define title, author, date

Document Structure: The Preamble

```
\documentclass[12pt, letterpaper]{article}  
  
\usepackage{graphicx}  
  
\title{My first LaTeX document}  
  
\author{Hubert Farnsworth\thanks{Funded by  
Overleaf.}}  
  
\date{August 2022}
```

Creating a Title

```
\documentclass[12pt, letterpaper]{article}  
\title{My first LaTeX document}  
\author{Hubert Farnsworth}  
\date{August 2022}
```

Creating a Title

```
\begin{document}
```

```
\maketitle
```

We have now added a title, author and date!

```
\end{document}
```

Creating a Title

My first LaTeX document

Hubert Farnsworth*

August 2022

We have now added a title, author and date to our first L^AT_EX document!

Adding Comments

```
\begin{document}
```

```
\maketitle
```

We have now added a title, author and date!

```
% This line is a comment. It won't be  
typeset.
```

```
\end{document}
```

Text Formatting

Some of the `\textbf{greatest}`
discoveries in `\underline{science}`
were made by `\textbf{\textit{accident}}`.

Text Formatting

Basic Commands:

- `\textbf{...}`: **Bold text**
- `\textit{...}`: *Italic text*
- `\underline{...}`: Underlined text
- `\emph{...}`: Emphasis (context-aware)

Adding Images

```
\documentclass{article}
```

```
\usepackage{graphicx}
```

```
\graphicspath{{images/}}
```


Adding Images

```
\begin{document}
```

The universe is immense...

```
\includegraphics{universe}
```

There's a picture of a galaxy above.

```
\end{document}
```

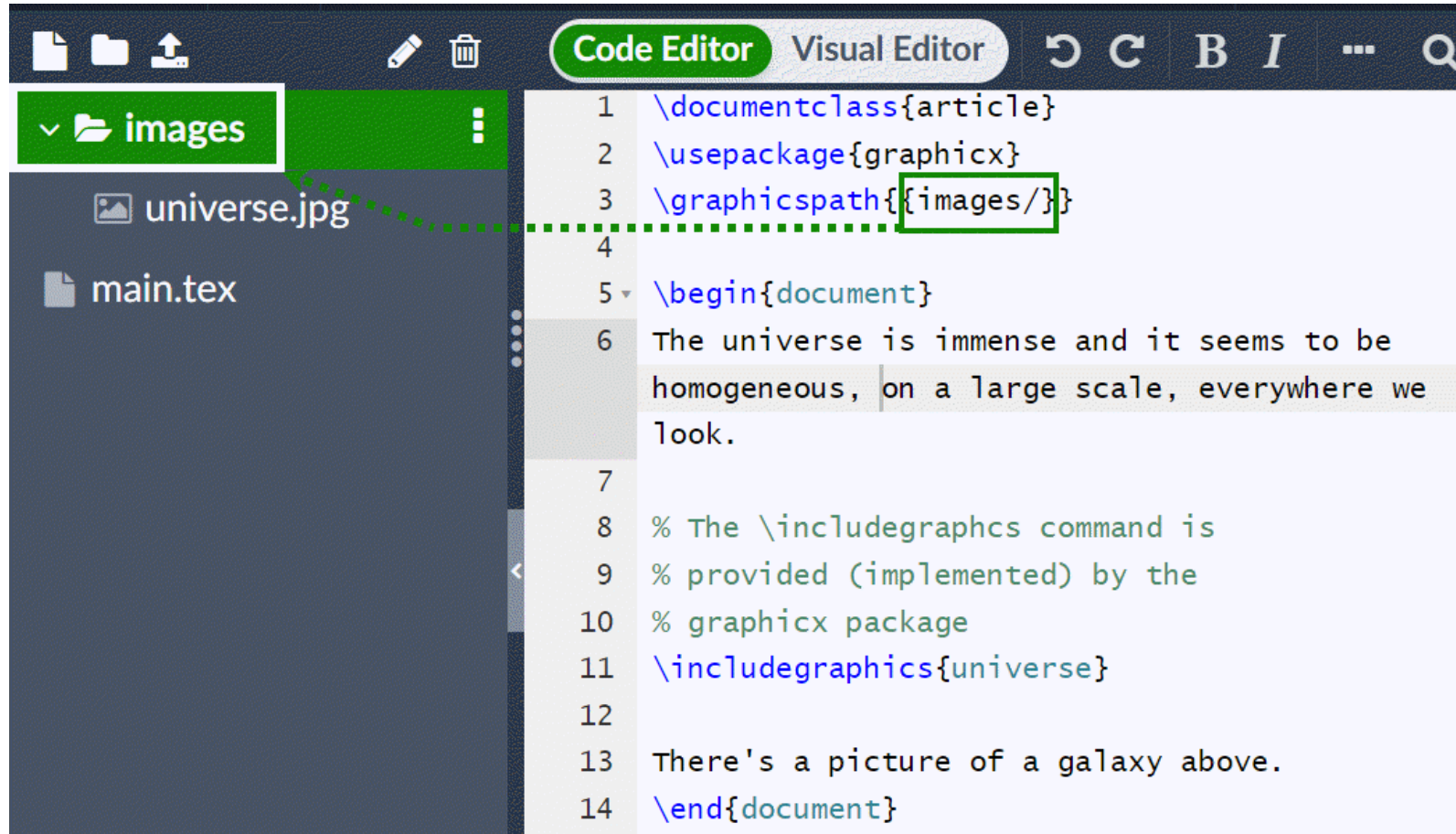
Adding Images

The universe is immense and it seems to be homogeneous, on a large scale, everywhere we look.



There's a picture of a galaxy above.

Adding Images



Adding Images

You need to use:

- `graphicx` package
- `\graphicspath{...}` command to specify directory
- `\includegraphics{filename}` to insert image

Captions and References

```
\begin{figure}[h]  
    \centering  
    \includegraphics[width=0.75\textwidth]  
{mesh}  
    \caption{A nice plot.}  
    \label{fig:mesh1}  
\end{figure}
```

Captions and References

As you can see in figure

`\ref{fig:mesh1}...`

This is on page `\pageref{fig:mesh1}`.

Captions and References

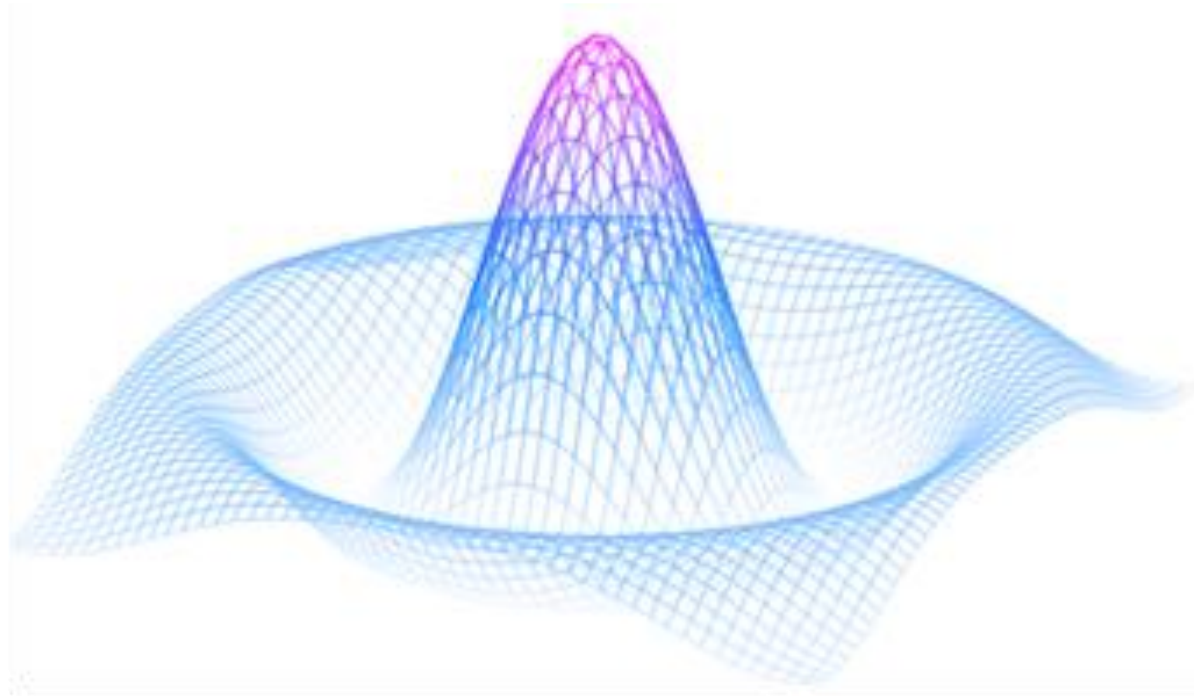


Figure 1: A nice plot.

As you can see in figure 1, the function grows near the origin. This example is on page 1.

Captions and References

Basic Commands

- `figure` environment: Wraps the image
- `\caption{...}`: Add caption
- `\label{...}`: Set label
- `\ref{...}`: Reference the figure

Unordered Lists

```
\begin{itemize}
```

```
    \item The individual entries are  
indicated with a black dot.
```

```
    \item The text may be of any length.
```

```
\end{itemize}
```

Ordered Lists

```
\begin{enumerate}
```

```
    \item This is the first entry in our  
list.
```

```
    \item The list numbers increase with  
each entry.
```

```
\end{enumerate}
```

Mathematics: Inline Mode

In physics, the mass-energy equivalence is stated

by the equation $E=mc^2$, discovered in 1905.

Mathematics: Inline Mode

Ways to write inline formulas:

- $\dots$

- $\left(\dots \right)$

- $\begin{math} \dots \end{math}$

Mathematics: Display Mode

The mass-energy equivalence is described
by $E=mc^2$

discovered in 1905 by Albert Einstein.

```
\begin{equation}
```

```
E=m
```

```
\end{equation}
```

Mathematics: Display Mode

Display mode:

- `\[ ... \]`: Unnumbered equation
- `\begin{equation} ... \end{equation}`:
Numbered equation

Advanced Mathematics

Examples

Subscripts: a_b and superscripts:
 a^b

$T^{i_1 i_2 \dots i_p}_{j_1 j_2 \dots j_q}$

Integrals and fractions:

$\int_0^1 \frac{dx}{e^x} = \frac{e-1}{e}$

Advanced Mathematics

Examples

Greek letters: ω δ
 Ω Δ

Operators: $\sin(\beta)$,
 $\cos(\alpha)$, $\log(x)$

Document Structure

```
\documentclass{book}
```

```
\begin{document}
```

```
\chapter{First Chapter}
```

```
\section{Introduction}
```

```
\subsection{First Subsection}
```

```
\section*{Unnumbered Section}
```

```
\end{document}
```

Document Structure

Sectioning Basic Commands:

- `\part`, `\chapter` (book, report classes)
- `\section`, `\subsection`, `\subsubsection`
- Add ``*`` to remove numbering

Creating Basic Tables

```
\begin{center}
```

```
\begin{tabular}{c c c}
```

```
cell1 & cell2 & cell3 \\\
```

```
cell4 & cell5 & cell6 \\\
```

```
cell7 & cell8 & cell9
```

```
\end{tabular}\end{center}
```

Tables with Borders

```
\begin{center} \begin{tabular}{|c|c|c|}  
  \hline  
  cell1 & cell2 & cell3 \\  
  cell4 & cell5 & cell6 \\  
  cell7 & cell8 & cell9 \\  
  \hline\end{tabular}\end{center}
```

Tables with Captions

```
\begin{table}[h!]
```

```
\centering \begin{tabular}{||c c c||}
```

```
\hline
```

```
Col1 & Col2 & Col3 \\\ [0.5ex]\hline\hline
```

```
1 & 6 & 787 \\\ \hline
```

```
\end{tabular}
```

Tables with Captions

```
\caption{Table to test captions.}
```

```
\label{table:data}
```

```
\end{table}
```

Table `\ref{table:data}` shows...

Automatic Table of Contents

```
\documentclass{article}
```

```
\title{Sections and Chapters}
```

```
\author{Your Name}
```

```
\date{\today}
```

Automatic Table of Contents

```
\begin{document}
```

```
\maketitle
```

```
\tableofcontents
```


Automatic Table of Contents

```
\section{Introduction}
```

This is the first section...

```
\section{Second Section}
```

More content here...

```
\end{document}
```

LaTeX Packages

- What is LaTeX Packages?

=> Packages extend LaTeX capabilities.

LaTeX Packages

- What is LaTeX Packages?

⇒ Packages extend LaTeX capabilities.

```
\usepackage[options]{packagename}
```

LaTeX Packages

Example:

```
\usepackage{graphicx} % Images
```

```
\usepackage{amsmath} % Advanced math
```

```
\usepackage[utf8]{inputenc}% UTF-8  
encoding
```

```
\usepackage{geometry} % Page layout
```

Abstracts

```
\documentclass{article}
```

```
\begin{document}
```

```
\begin{abstract}
```

This is a simple paragraph at the beginning. A brief introduction about the main subject.

```
\end{abstract}
```

Paragraphs and Line Breaks

- **Create new paragraph:** Press Enter twice (blank line)
- New line in same paragraph: `\\` or `\newline`

Paragraphs and Line Breaks

- **Create new paragraph:** Press Enter twice (blank line)
- New line in same paragraph: `\\` or `\newline`
- **Note:** Don't abuse `\\` for spacing!

Paragraphs and Line Breaks

First paragraph content here.

Second paragraph starts here.

Same paragraph but `\\` new line here.

Tips and Best Practices

DO:

- Use appropriate document classes
- Load necessary packages in preamble
- Use `\label` and `\ref` for cross-references
- Comment your code for maintainability

Tips and Best Practices

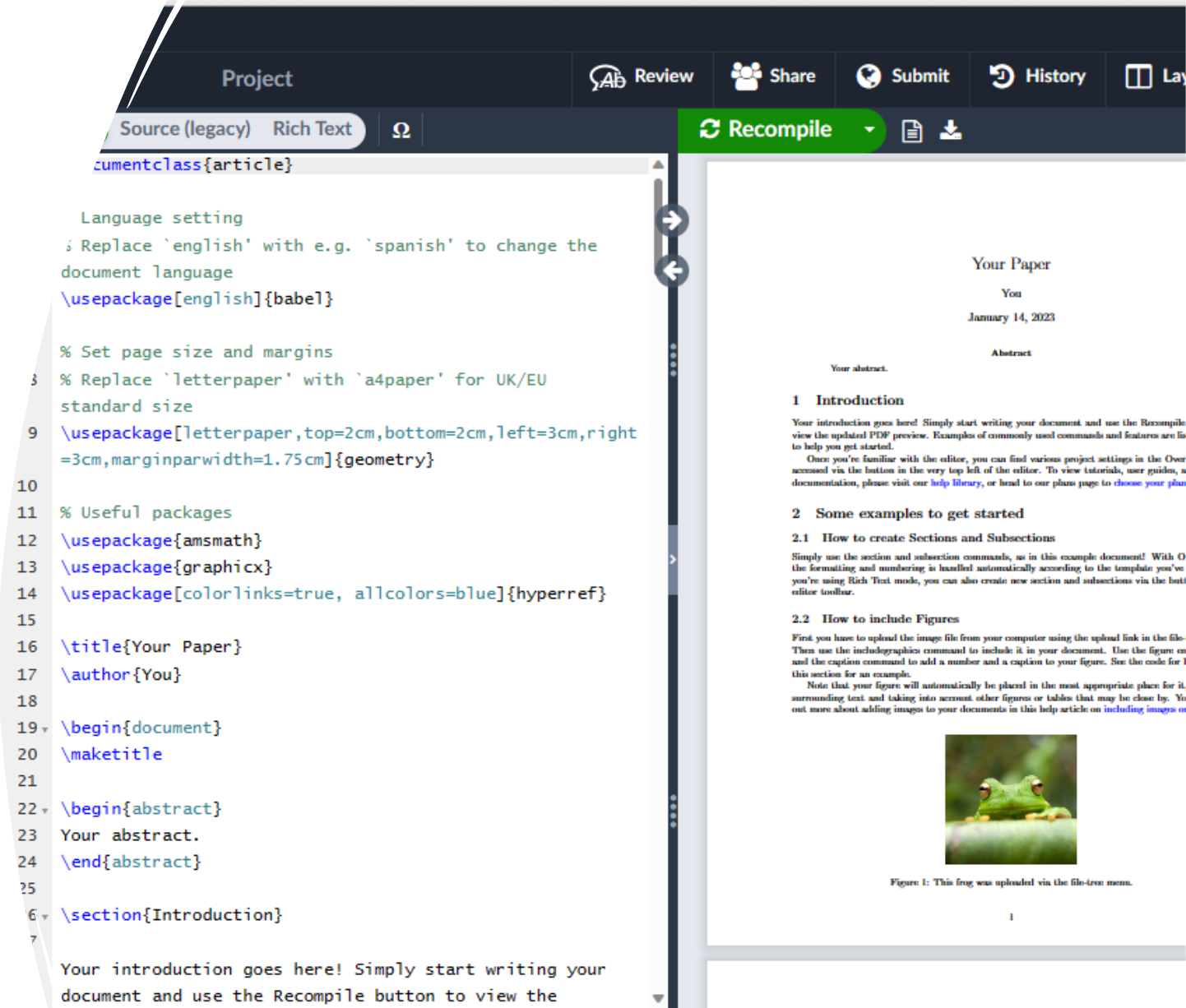
DONOT:

- Abuse `\` for paragraph spacing
- Add extensions to image filenames in `\includegraphics`
- Use `$$` for display math (use `\[...]` instead)

Learning Resources

Overleaf:

- Gallery with thousands of templates
- Detailed documentation
- Visual Editor and Code Editor



The screenshot displays the Overleaf online LaTeX editor interface. The top navigation bar includes buttons for 'Project', 'Review', 'Share', 'Submit', 'History', and 'Layers'. Below this, a toolbar shows 'Source (legacy)', 'Rich Text', and a 'Recompile' button. The main area is split into two panes. The left pane is the code editor, showing LaTeX source code for a document class 'article'. The code includes language settings, package loading (babel, geometry, amsmath, graphicx, hyperref), title and author information, and the start of an abstract and introduction section. The right pane is the PDF preview, showing a document titled 'Your Paper' with a date of 'January 14, 2023'. It includes an abstract section and an introduction section with a paragraph of text. A small image of a green frog is visible in the introduction section, with a caption below it: 'Figure 1: This frog was uploaded via the file-tree menu.'

```
documentclass{article}

% Language setting
% Replace 'english' with e.g. 'spanish' to change the
document language
\usepackage[english]{babel}

% Set page size and margins
% Replace 'letterpaper' with 'a4paper' for UK/EU
standard size
\usepackage[letterpaper,top=2cm,bottom=2cm,left=3cm,right=
=3cm,marginparwidth=1.75cm]{geometry}

% Useful packages
\usepackage{amsmath}
\usepackage{graphicx}
\usepackage[colorlinks=true, allcolors=blue]{hyperref}

\title{Your Paper}
\author{You}

\begin{document}
\maketitle

\begin{abstract}
Your abstract.
\end{abstract}

\section{Introduction}

Your introduction goes here! Simply start writing your
document and use the Recompile button to view the
```

CTAN Comprehensive T_EX Archive Network

Learning Resources

Deckblatt Hochladen Browsen

Ort: CTAN Pakete oberdiek

CTAN

oberdiek – A bundle of packages submitted

- Search packages by topic

The bundle comprises packages to provide:

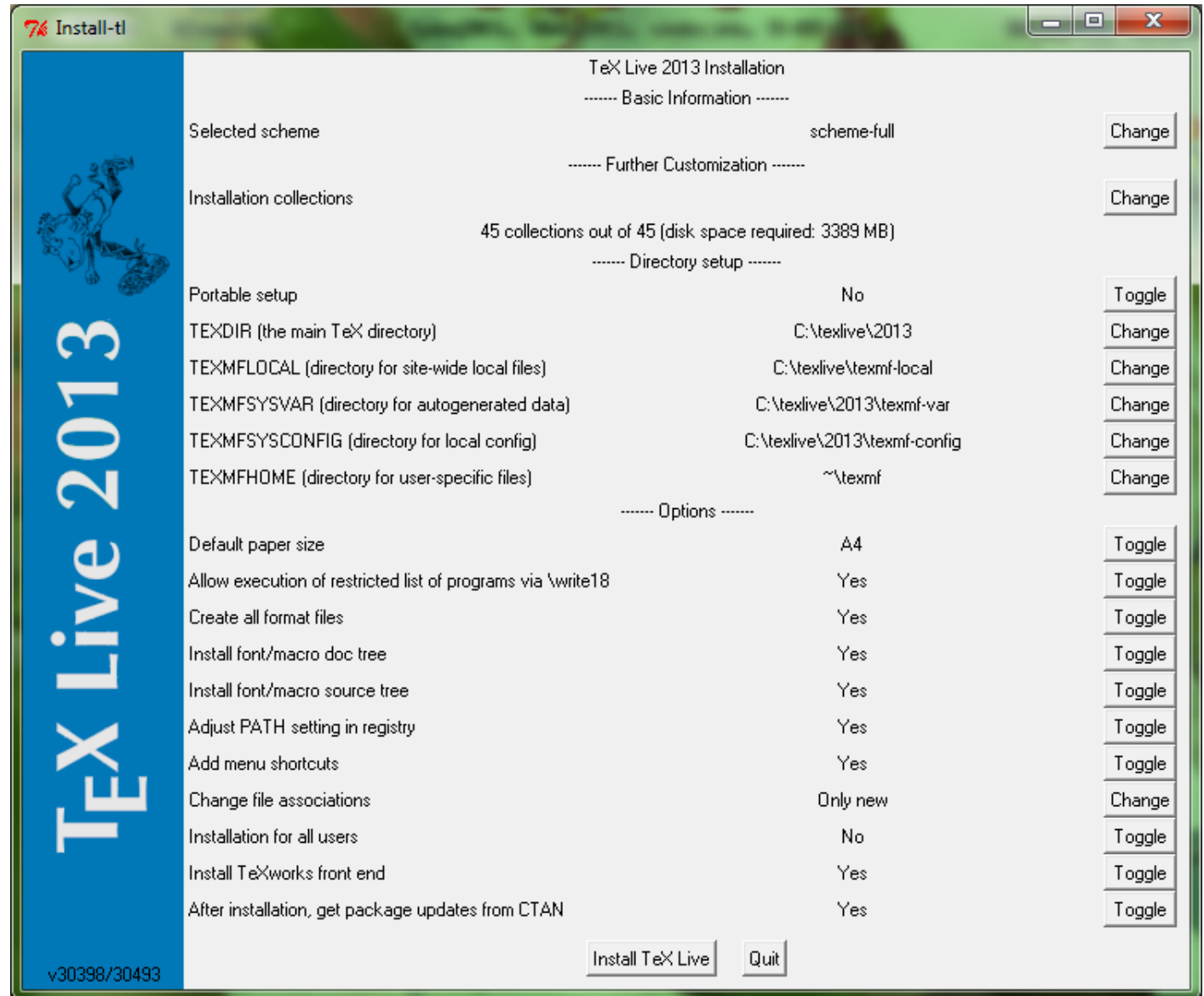
- [aliascnt](#): 'alias counters';
- [bmpsize](#): get bitmap size and resolution data;
- [centernot](#): a horizontally-centred `\not` symbol;
- [chemarr](#): extensible chemists' reaction arrows;
- [classlist](#): record information about document class(es) used;
- [colonequals](#): poor man's mathematical relation symbols;

- Documentation for each package

Learning Resources

TeX Live:

- Complete LaTeX installation
- Annual updates





QA